

Assignment - 1 Computer network

Discuss the role of Computer network in education and e-learning system.

Computer networks play a vital role in modern education and e-learning system by enabling communication, collaboration, and access to digital resources across the world.

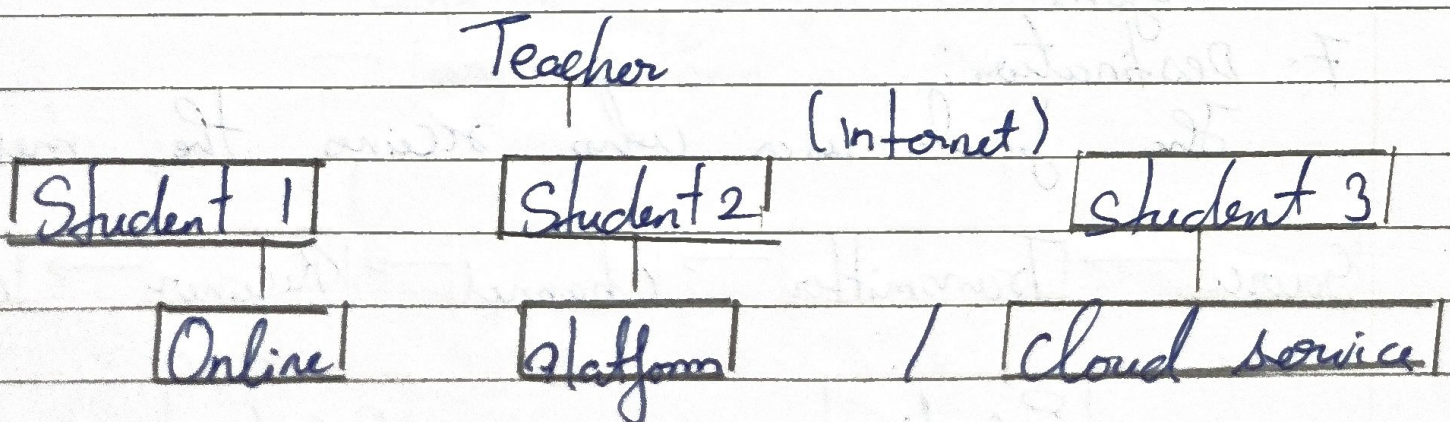
i. Online learning platform:
network enable access to platform like Coursera, udemy and Khan Academy where student can attend courses remotely.

ii. Virtual classroom
Application such as Zoom & Google Meet allow live classes, webinars, and doubt-clearing sessions.

iii. Resource sharing:
Teacher and student can share notes, PDFs, video, and assignment through cloud services like google drive.

iv. Collaboration:
Student can work on group projects using shared document and communication tools.

v. Distance Education:
Student in remote area can access quality education without physically attending institutions.



2. Elaborate a Communication system. Explain its basic components with a neat diagram.

— A Communication system is a system that transfer information from one place to another through a transmission medium.

A Communication system consists of device and methods used to transmit data from sender to receiver, accurately and efficiently.

Basic Component of Communication System.

1. Sender (Source):
The device that generates and sends the message.

2. Input Transducer:
Convert physical message (voice, image) into electrical signals.

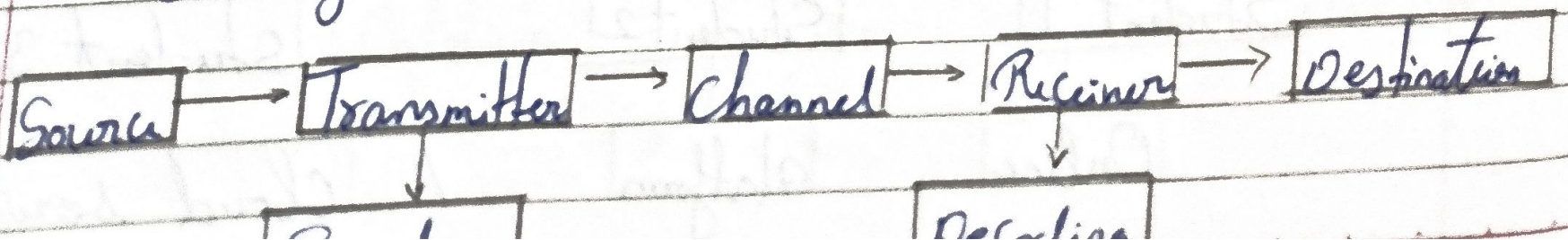
3. Transmitter:
process and encodes the signal for transmission.

4. Transmission Medium (Channel):
The path through which signal travel (wired or wireless)

5. Receiver:
Receives and decodes the signal

6. Output transducer:
Converts electrical signal back into original form.

7. Destination:
The final user who receives the message.



3. classify Computer network based on geographical area (LAN, MAN, WAN) and explain each with examples.

- Computer network are classified based on geographical Coverage into LAN, MAN & WAN.

1. LAN: (Local area network)

A LAN covers a small geographical area like a room, office, school or building.

Characteristics:

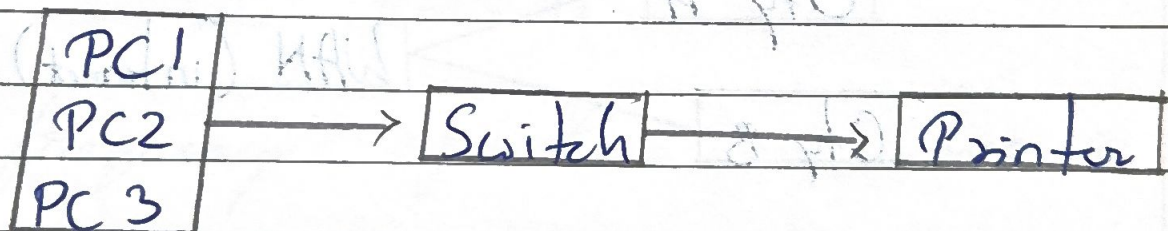
- High speed
- Low Cost
- Privately owned

Example:

Computer lab network
Office network

Example: of LAN technology: Ethernet.

Diagram:



2. MAN (Metropolitan Area Network)

A MAN covers a city or metropolitan area.

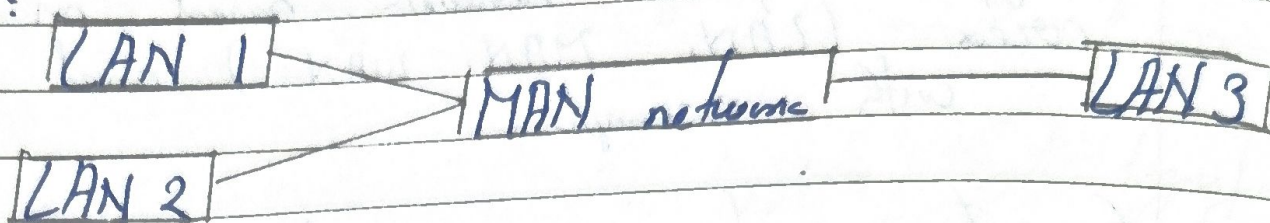
Characteristics:

- Medium Speed
- Connects multiple LANs
- Larger than LAN

Example:

City Cable TV network
University Campus network

Diagram:



3. WAN (wide area network)

A WAN covers a large geographical area such as a country or world.

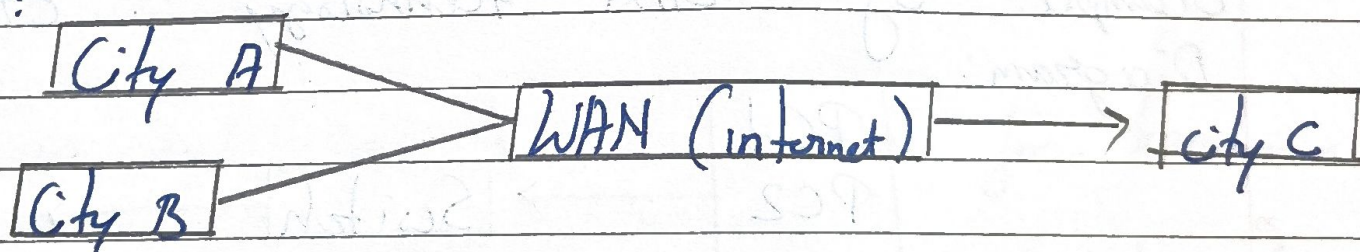
Characteristics:

- Lower speed compared to LAN
- Expensive
- Public or private ownership

Example:

- The internet
- Bank network across countries

Diagram:



4. A digital system transmits data at 9600 bps. How long will it take to send a file of size 48000 bits?

Given : Data rate = 9600 bps
File Size = 48000 bits

Formula: $\text{Time} = \frac{\text{File Size}}{\text{Data Rate}}$

Calculation: $\text{Time} = \frac{48000}{9600}$

Time = 5 Sec

$\therefore$ It will take 5 Second to transmit the file.

5. A Sine wave Complete 2000 cycle in 0.5 sec
Determine its frequency.

- Given: Number of Cycle = 2000
Time = 0.5 second

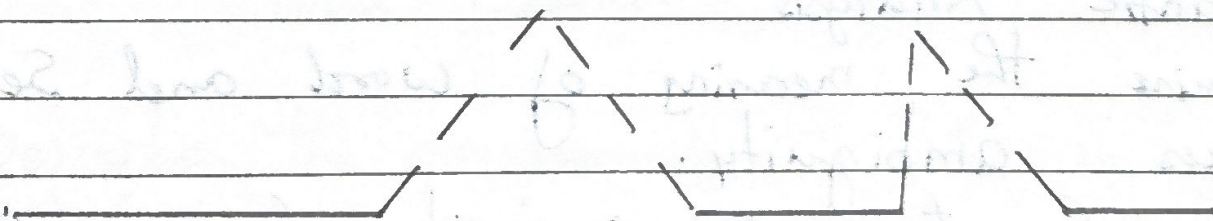
Formula: Frequency (f) = $\frac{\text{Number of Cycle}}{\text{Time}}$

Calculation: $f = \frac{2000}{0.5}$

$\therefore f = 4000 \text{ Hz}$

$\therefore$ The frequency of the sine wave is 4000 Hz.

Diagram of sine wave:



Frequency represent how many cycles are Completed in one second.